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Android Ransomware-like Activity Early Warning in a Controlled Lab

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Problem Description

Traditional antivirus tools may fail to detect unknown ransomware, while rapid file modifications can cause data loss before detection.

Key Issues:
Unknown Threats: Signature-based tools may miss new ransomware.
Rapid File Changes: Multiple files may be rewritten or renamed within seconds.
Delayed Alerts: Late detection increases potential file damage.
Android Restrictions: App sandboxing limits cross-app monitoring.
Safe Testing: Detection must be evaluated using harmless simulations in a controlled lab.

Objective: Develop a behaviour-based detector that identifies suspicious file activity in a designated test directory, raises early alerts, and stops only the controlled test workload.

Background

Android File Security: Android applications store and modify files within designated storage areas.
Ransomware-Like Behaviour: Rapid file rewriting and renaming can indicate suspicious activity that may threaten data integrity.
Traditional Detection: Signature-based methods may not identify previously unknown threats.
Behaviour-Based Monitoring: Tracking file activity patterns can help identify suspicious behaviour without relying solely on known malware signatures.
Controlled Testing: An Android emulator or rooted lab device provides an environment to evaluate detection using harmless simulations.
Optimization Focus: Improve early warning by reducing detection delay and the number of files modified before an alert, while maintaining detection accuracy.

Literature & RFC Review

Android Ransomware Research: Behaviour-based detection identifies suspicious app activity beyond known malware signatures.
NIST SP 1800-26: Provides guidance on detecting and responding to data-integrity attacks.
Android Storage Security: Scoped storage restricts file access, supporting monitoring of explicitly permitted directories.
RFC 5424 (Syslog): Provides a standard format for recording security events and alerts.
Research Gap: Early detection of rapid file rewriting and renaming, evaluated using detection delay, files changed before alert, precision, and recall in a controlled Android lab.
References: NIST SP 1800-26 · Android Developers — Storage · IETF RFC 5424

Proposed Solution

Develop a **behaviour-based early-warning system** to detect ransomware-like file activity in a controlled Android environment.
File Monitoring: Monitor file creation, rewriting, and renaming in a designated test directory.
Behaviour Analysis: Identify suspicious patterns such as rapid, repeated modifications across multiple files.
Early-Warning Alerts: Trigger an alert when file activity exceeds defined behavioural thresholds.
Controlled Response: Stop only the test workload when suspicious activity is detected.
Performance Evaluation: Test using two benign workloads and three harmless ransomware-like simulations.
Key Metrics: Detection delay · Files changed before alert · Precision · Recall
Expected Outcome: Early identification of suspicious file activity while minimizing false alerts and protecting the scope of the controlled test environment.

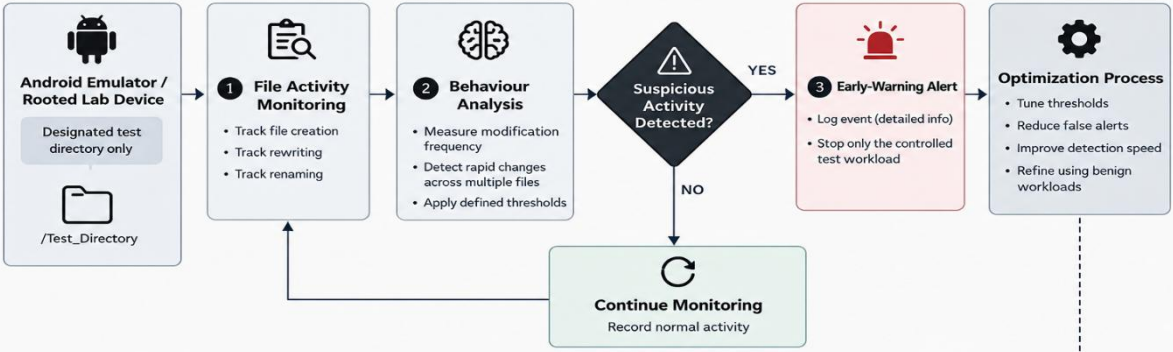
Optimization Proposed

Parameter	Before Optimization	After Optimization
File Monitoring	Basic file tracking	Monitor designated test directory
Detection	Simple rule-based checks	Behaviour-based threshold detection
Alert Speed	Potentially delayed alerts	Early warning for rapid file changes
Accuracy	Untuned thresholds	Thresholds refined using benign workloads
Response	Manual intervention	Alert and stop controlled test workload
Evaluation	Limited performance analysis	Measure delay, files changed, precision and recall

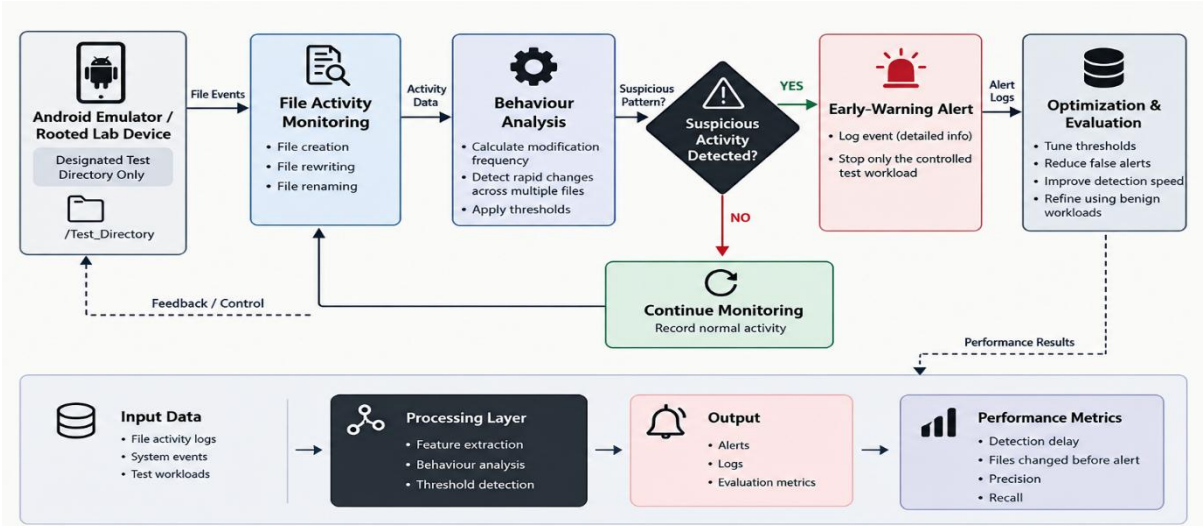
Expected Outcome

- Faster detection
- Fewer false alerts
- Reduced file changes before warning
- Controlled Android environment

Solution Architecture & Design



Data Flow Diagram



Deployment Timeline

Phase 1: Setup
Configure Android emulator and test directory.
Prepare benign workloads and harmless simulations.
Phase 2: Development
Implement file monitoring and behaviour analysis.
Configure detection thresholds and early alerts.
Phase 3: Testing & Optimization
Test 2 benign workloads and 3 simulations.
Measure detection delay, files changed, precision, and recall.
Refine thresholds to reduce false alerts.
Phase 4: Validation
Validate detection performance.
Demonstrate alerts and controlled workload termination.
Document results and finalize the prototype.
Methodology: Setup → Develop → Test → Optimize → Validate

Bibliography I

1.Android Developers — *Data and File Storage Overview.*
Official Android documentation.
2.NIST — *SP 1800-26: Detecting and Responding to Ransomware and Other Destructive Events.*
3.IETF — *RFC 5424: The Syslog Protocol.*
4.MITRE ATT&CK — *Mobile Threat Techniques.*
5.Research Papers — Studies on Android ransomware detection and behaviour-based malware analysis.

Thank You for Your Attention!